RAI-818: INTELLIGENT MOBILE ROBOTICS

- Configuration Space (Cont …)
- Topology

Text : Principles of Robot Motion – Theory, Algorithms and Implementations

H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

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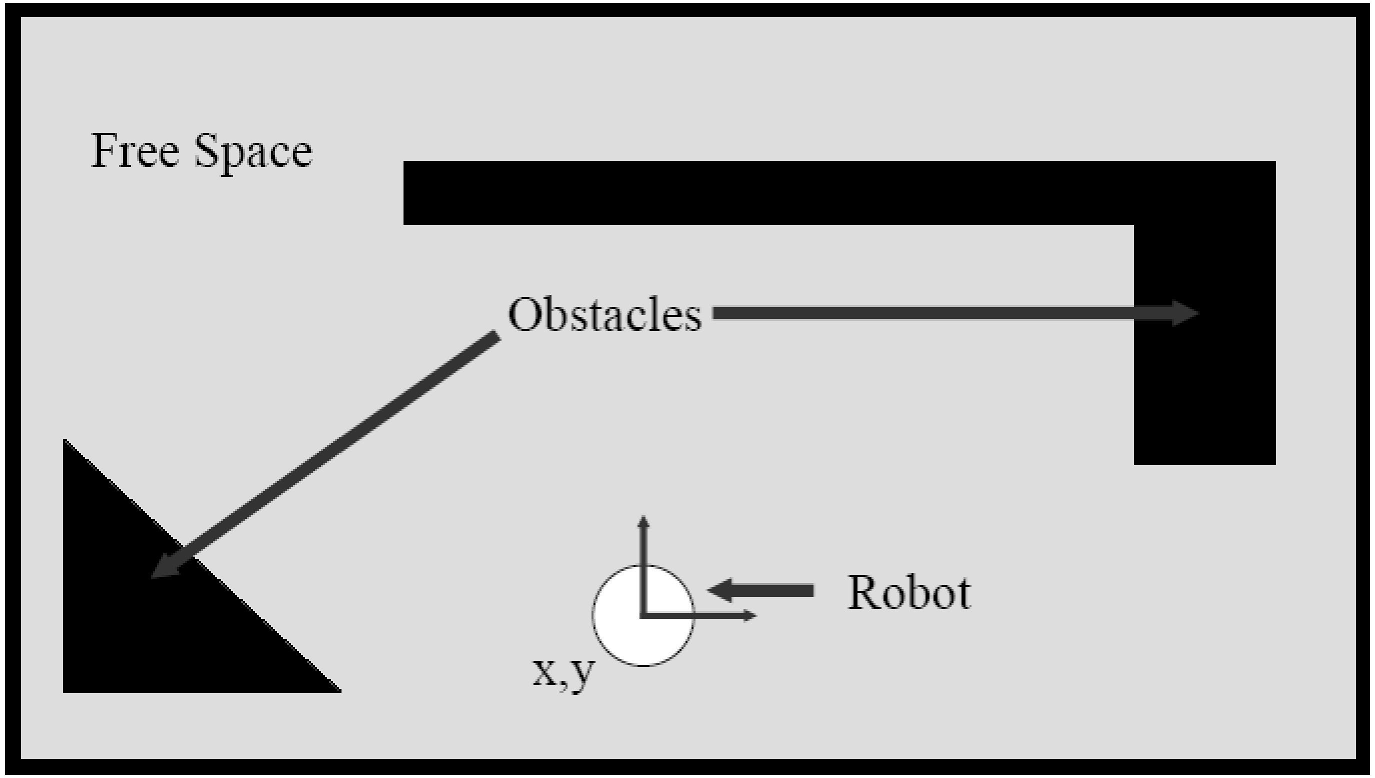
Representing Obstacles in C-Space

Example of a World (and Robot)

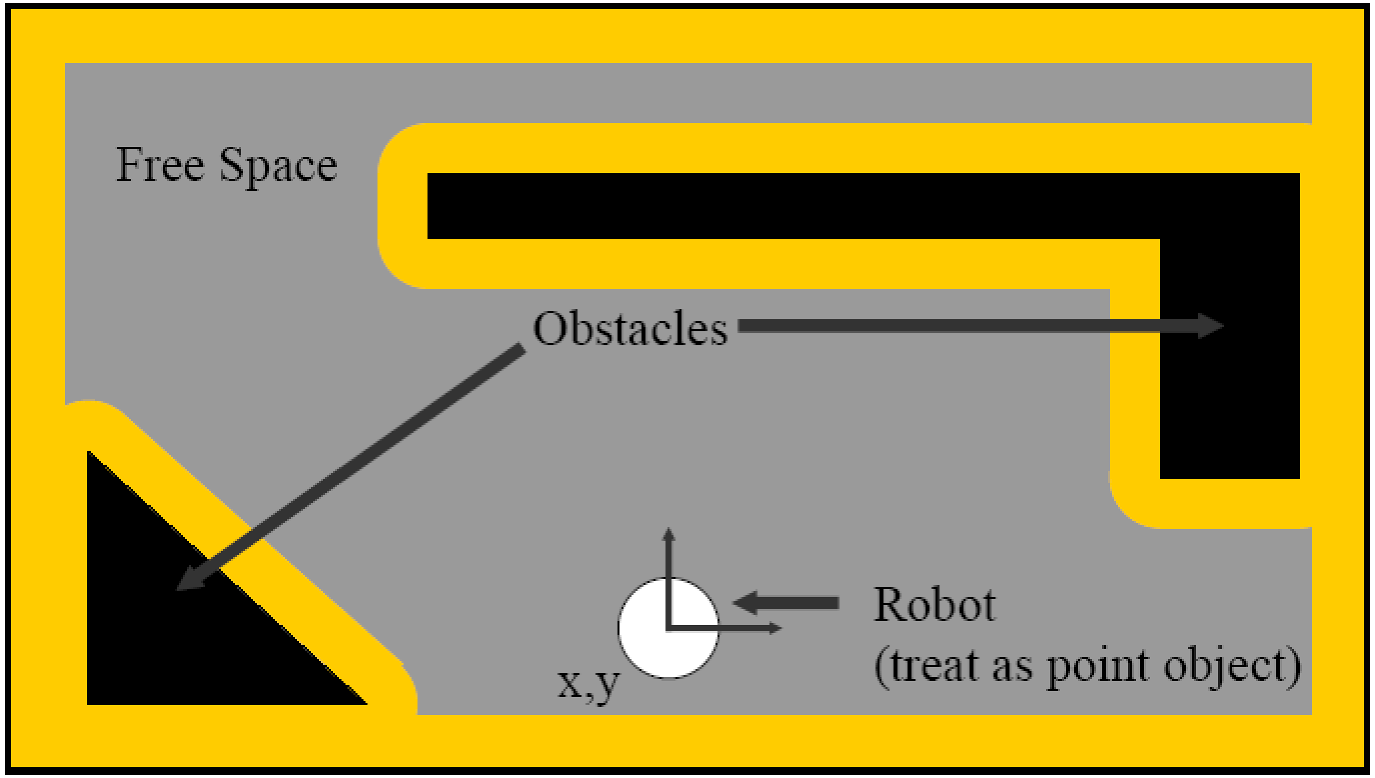


Configuration Space

Defining futures

Representing Obstacles in C-Space

Configuration Space: Accommodate Robot Size

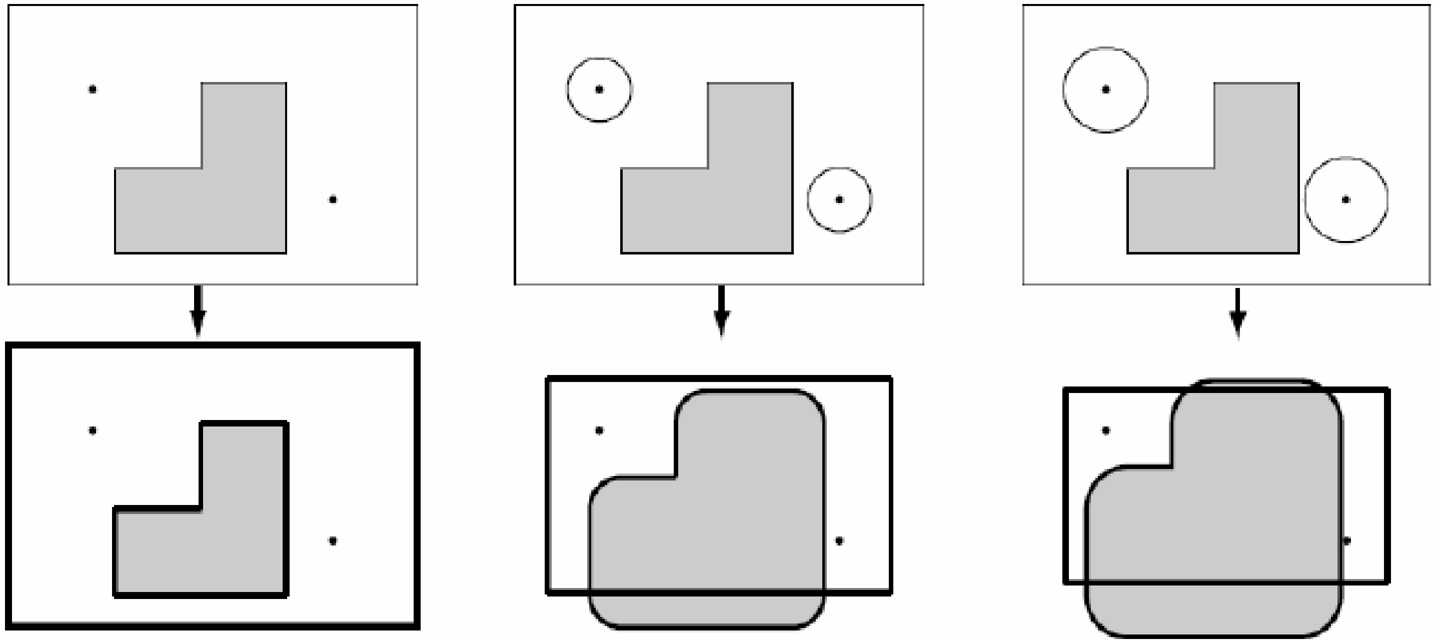


Representing Obstacles in C-Space

Trace Boundary of Workspace

Workspace

C-Space

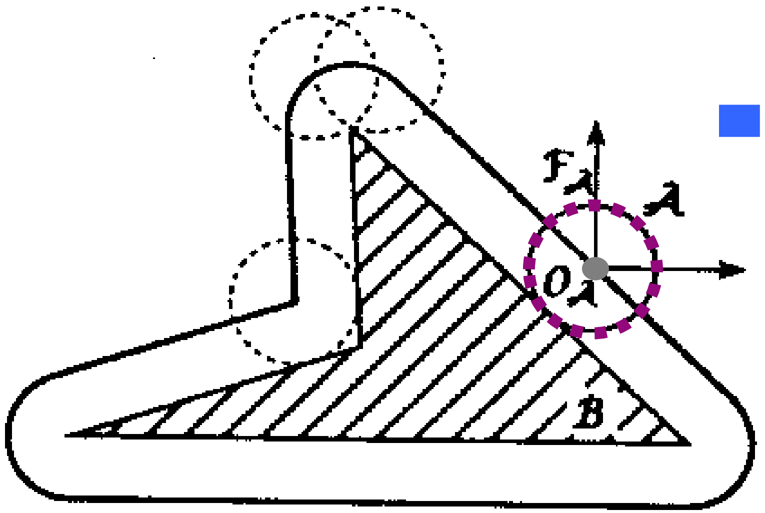


$$QO_i = \{q \in Q \mid R(q) \cap WO_i \neq 0\}$$

Representing Obstacles in C-Space

Disc in 2D workspace

workspace

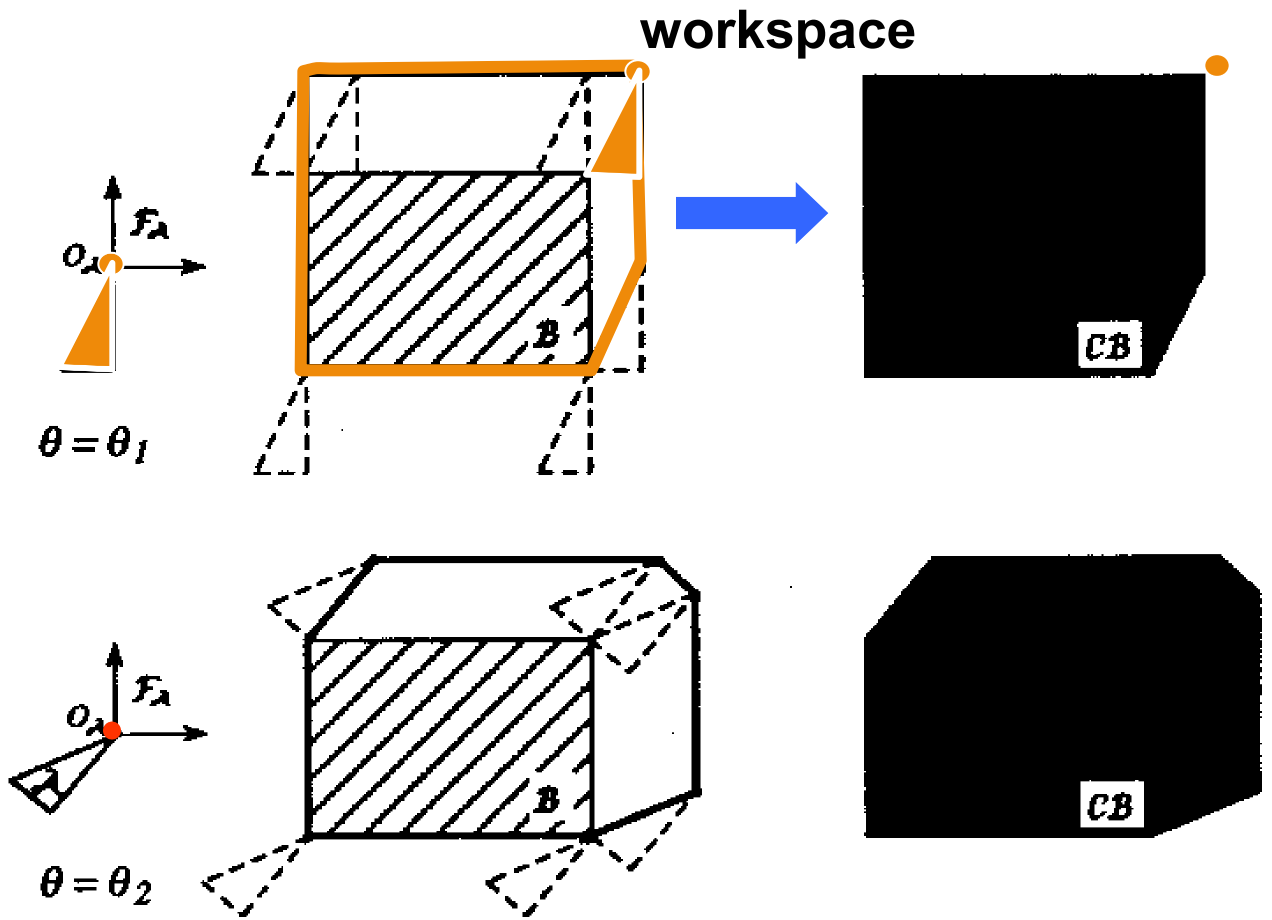


Defining futures

- **C- Space**
  - Non-Point Robots
  - What's a C-Space
  - Examples
  - Obstacle in C-Space
  - C-Space Dimensions
  - Constraints
    - Non-Holonomic
  - **Representing Obstacles**

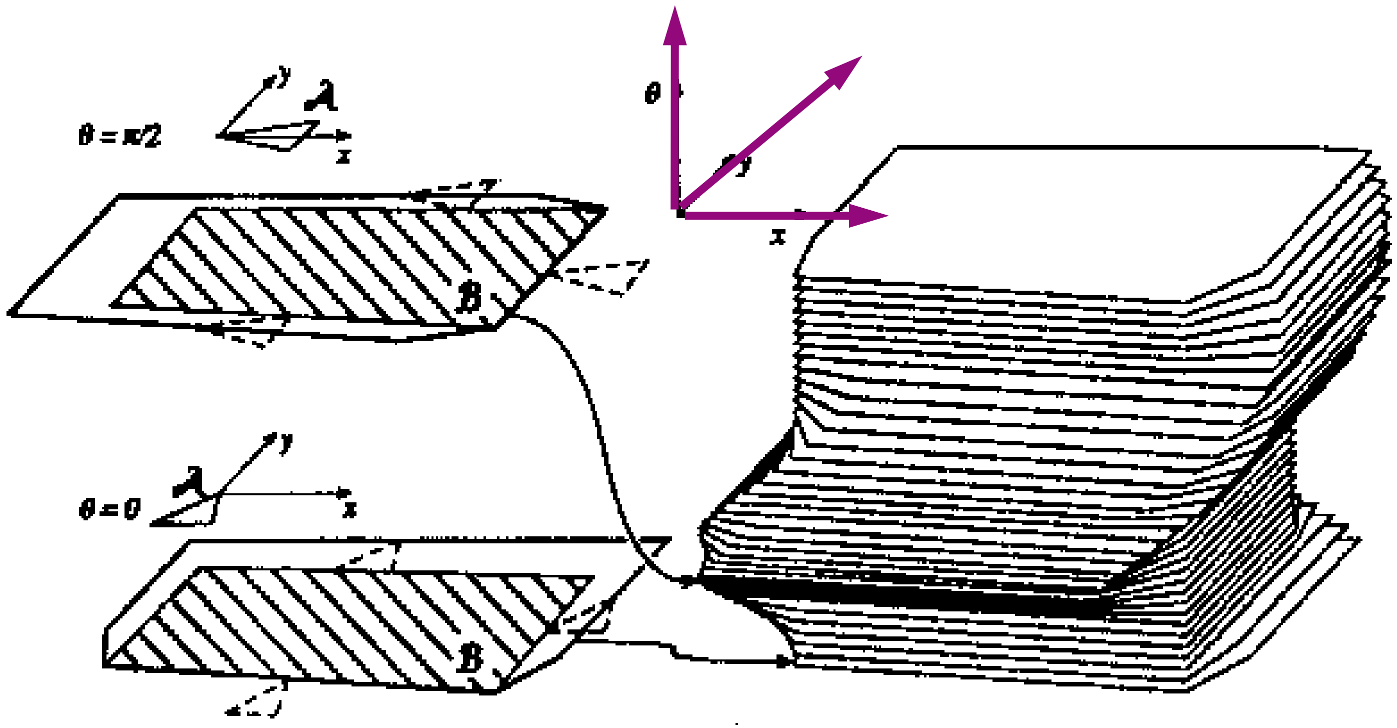
Representing Obstacles in C-Space

Polygonal Robot Translating in 2D Workspace



Representing Obstacles in C-Space

Polygonal Robot Translating in 2D Workspace



Defining futures

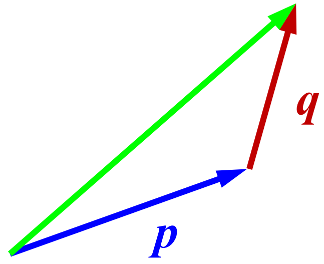
Configuration Space

Defining futures

Minkowski Sum

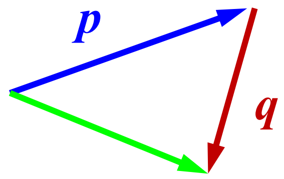
- The *Minkowski sum* of two sets P and Q, denoted by $P \oplus Q$, is defined as

$$P \oplus Q = \{ p+q \mid p \in P, q \in Q \}$$



- Similarly, the *Minkowski difference* is defined as

$$P \ominus Q = \{ p-q \mid p \in P, q \in Q \}$$



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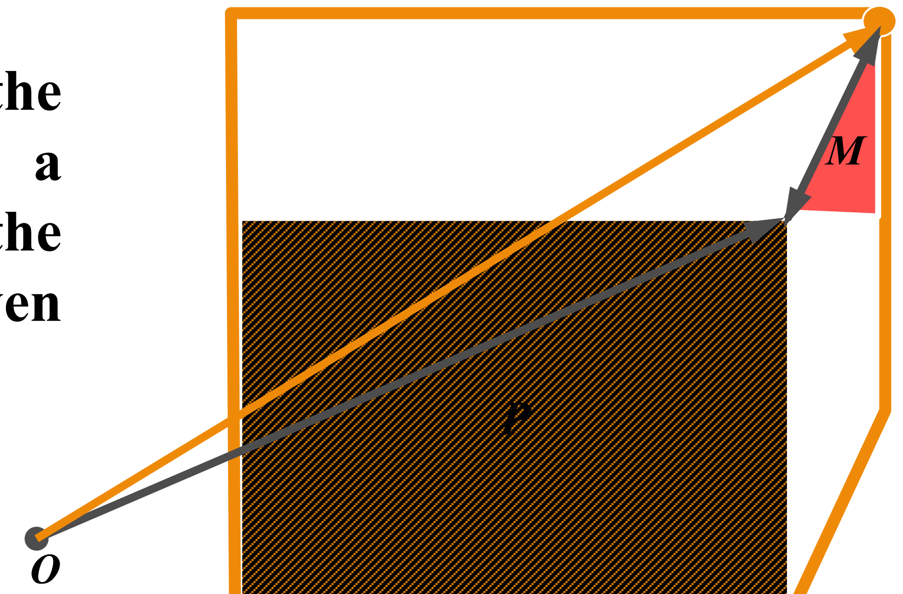
Minkowski Sum

- The *Minkowski sum* of two convex polygons P and Q of m and n vertices respectively is a convex polygon $P + Q$ of $m + n$ vertices
  - The vertices of $P + Q$ are the *sums* of vertices of P and Q

If P is an obstacle in the workspace and M is a moving object. Then the C-space obstacle is given by $P - M$



$$\Rightarrow P \ominus M = \{p - m \mid p \in P, m \in M\}$$

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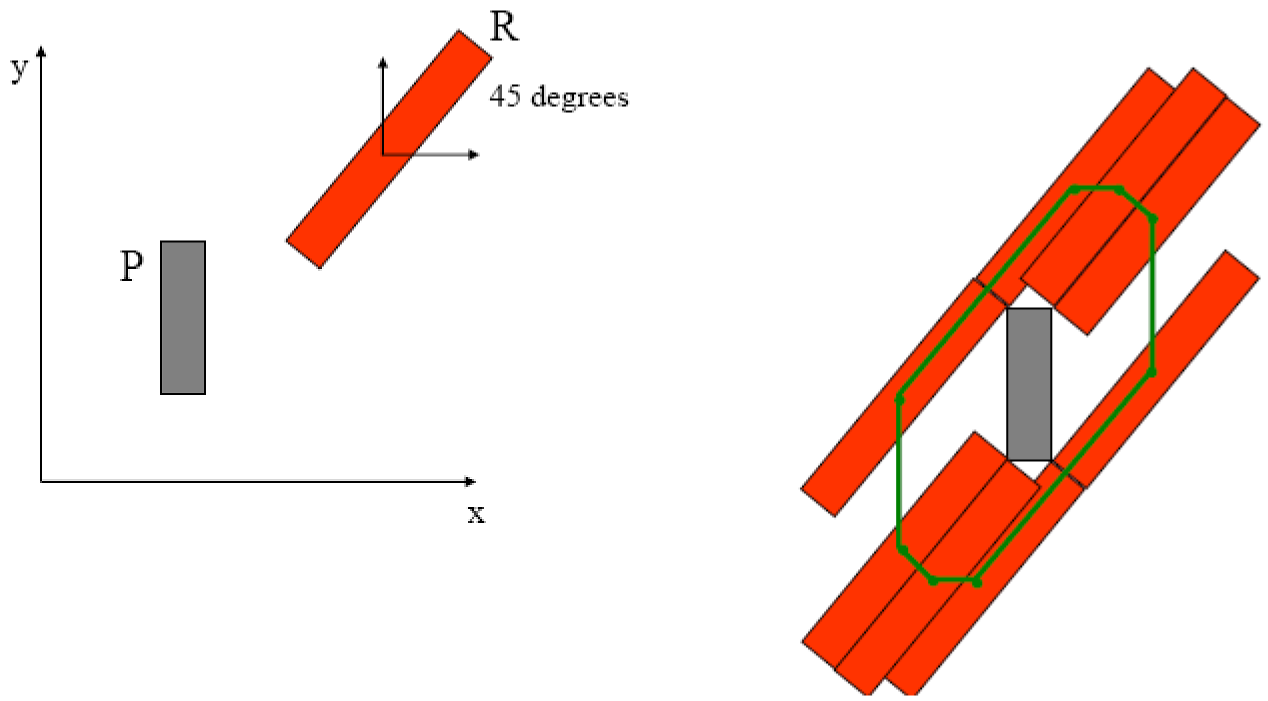
    + **Minkowski Sum**

Defining futures

Minkowski Sum

Taking the cross section of configuration space in which the robot is rotated 45 degrees...



How many sides does P ⊕R have?

Configuration Space

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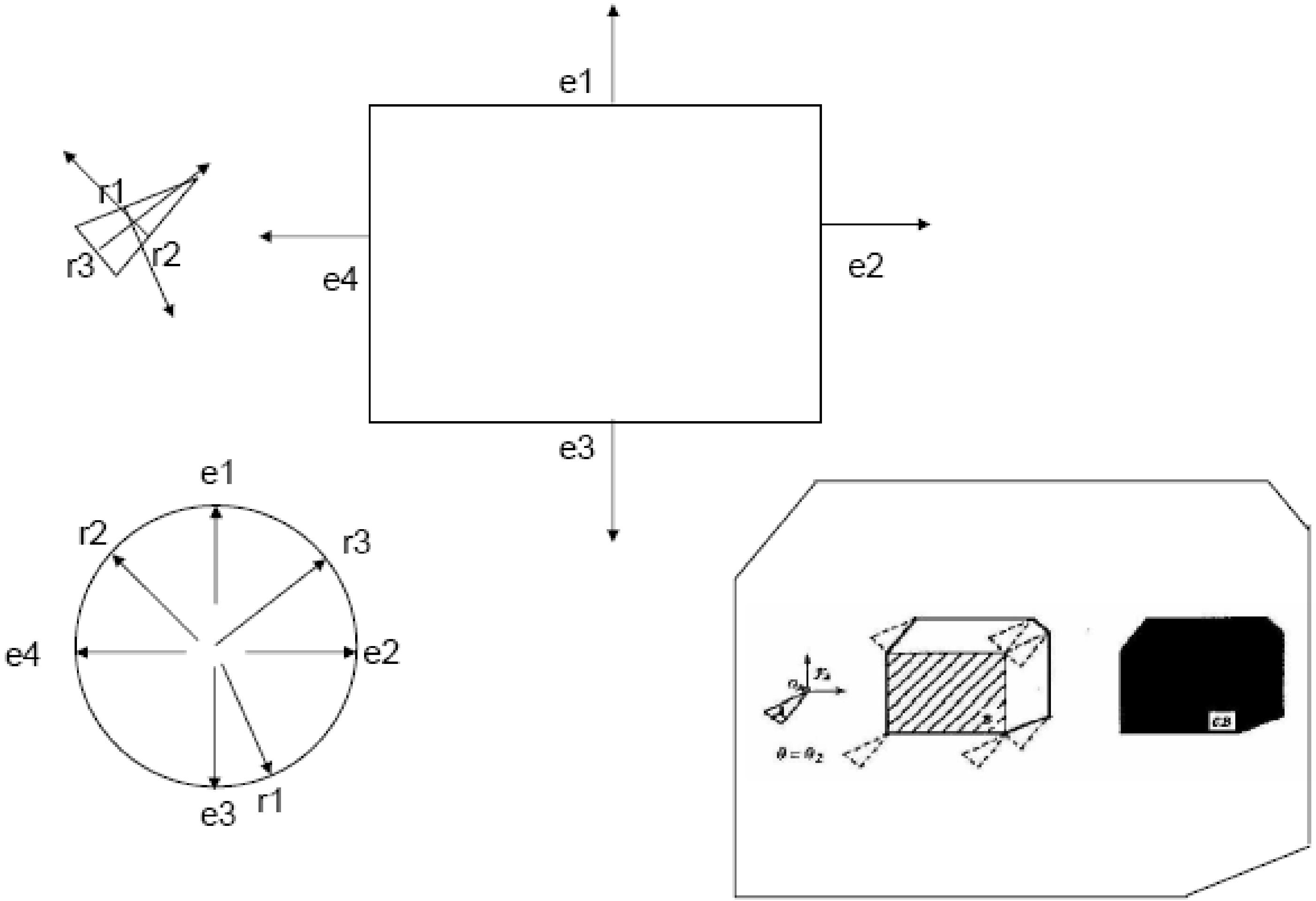
Minkowski Sum – Star Algorithm



Configuration Space

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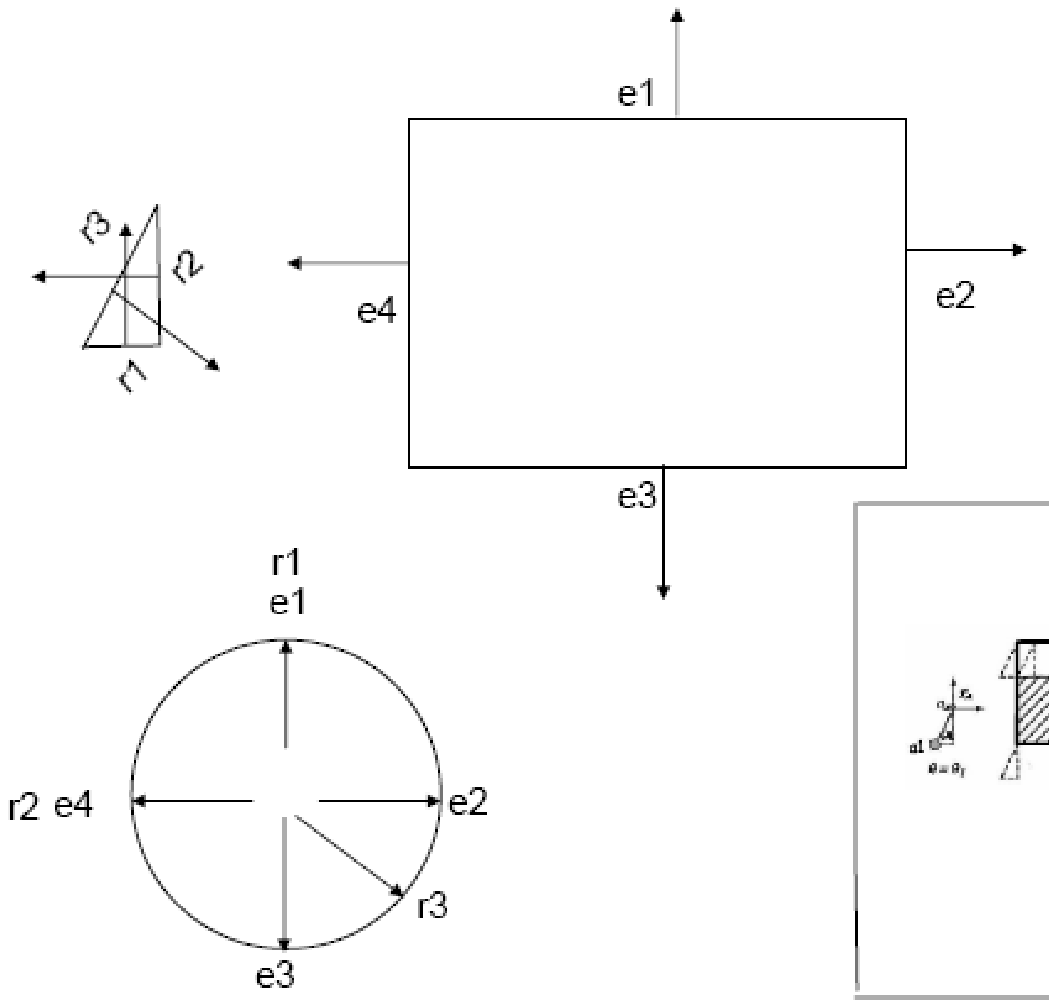
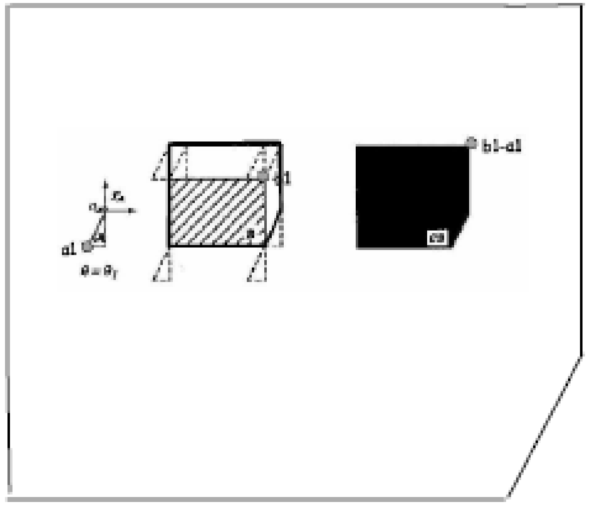
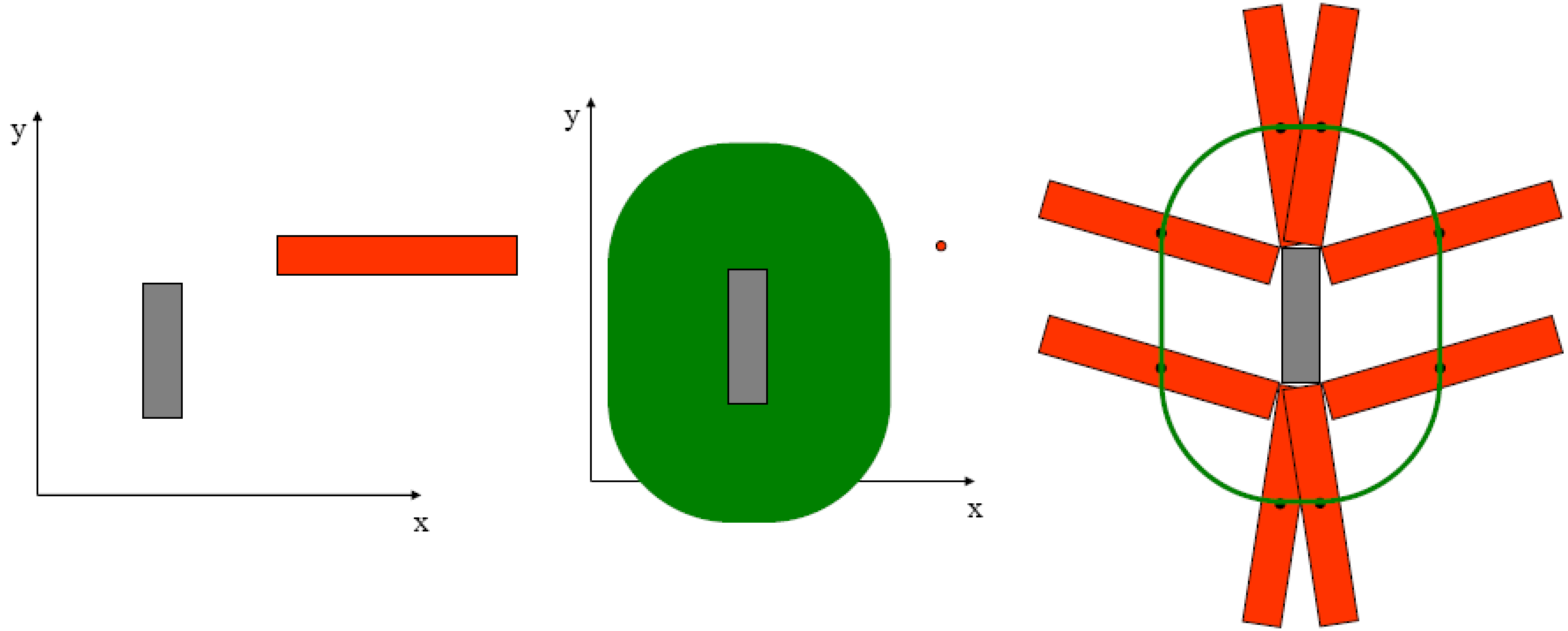
Minkowski Sum – Star Algorithm





Minkowski Sum – Additional Dimensions

What would the configuration space of a rectangular robot (*red*) in this world look like? Assume it can *translate* and *rotate* in the plane. (The *blue* rectangle is an *obstacle*.)



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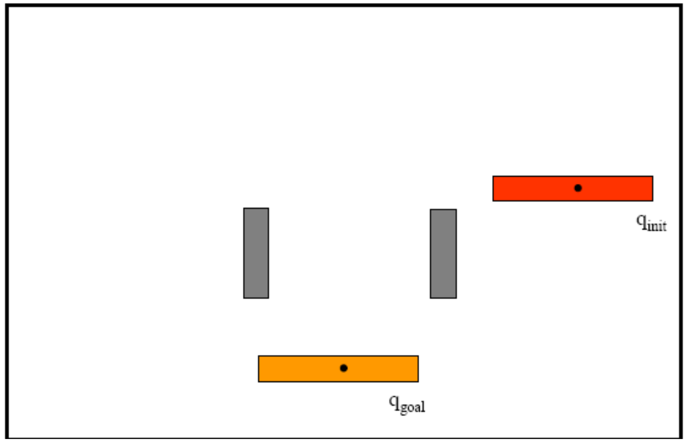
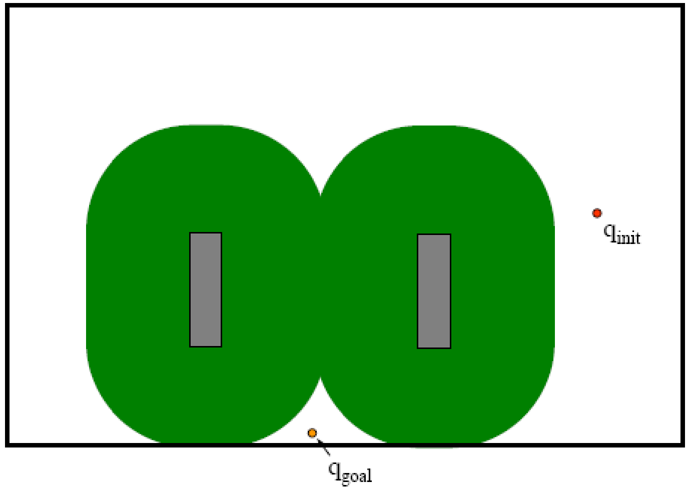
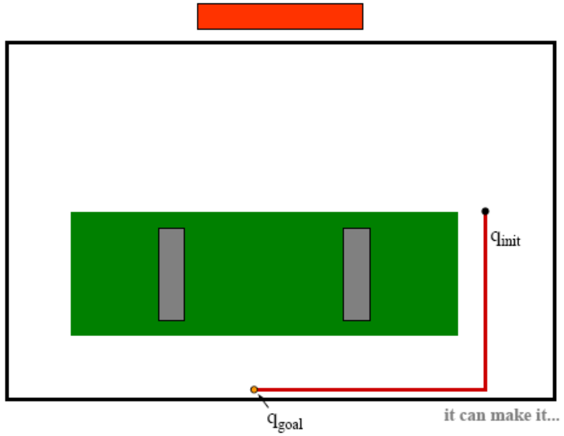
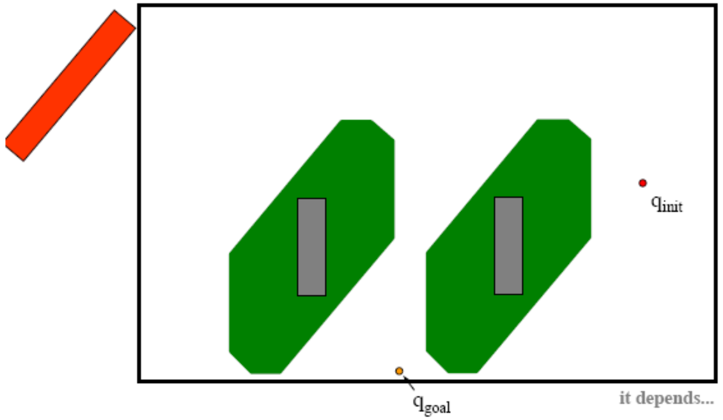
Minkowski Sum – A problem





But if it moves without rotating





Configuration Space

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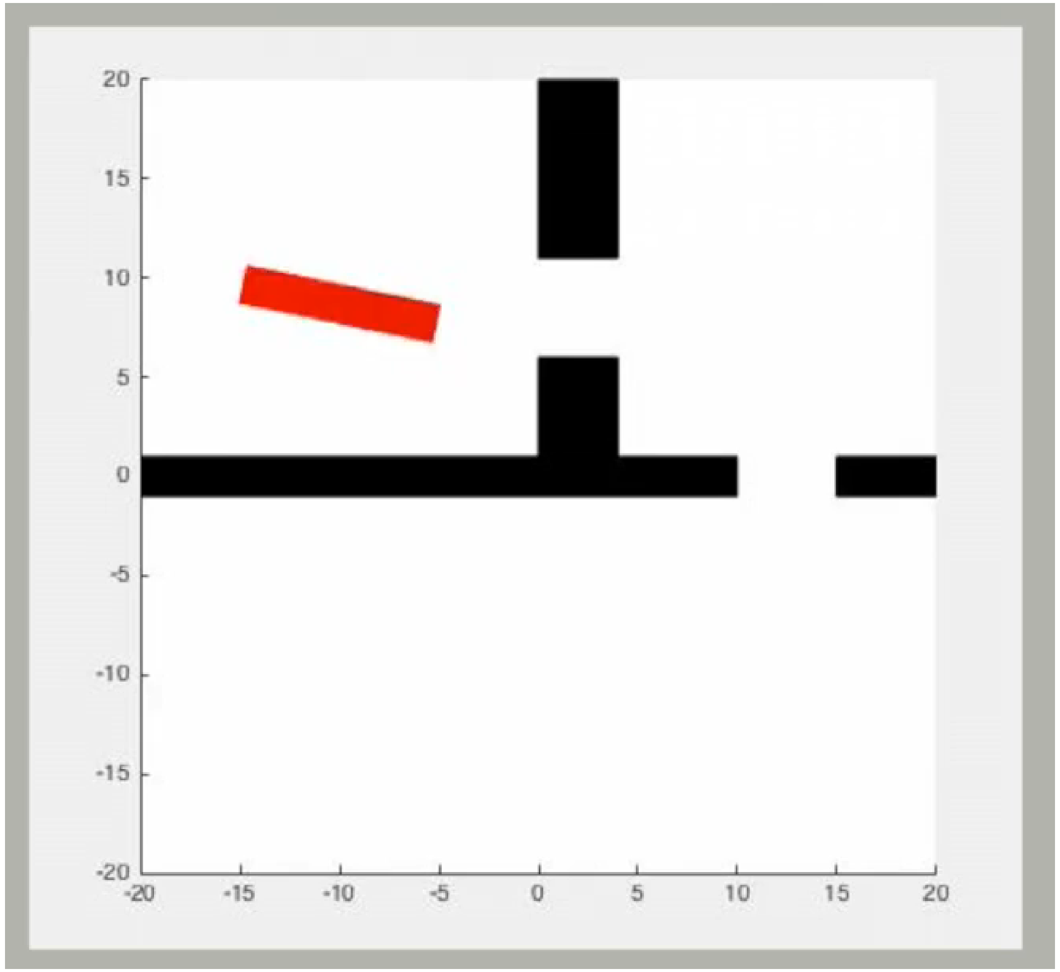
Minkowski Sum – Complex Example

Rotating and Translating Robot



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Configuration Space

Topology

- Branch of mathematics that deals with spatial relationships that are preserved under *bi-continuous deformation* (*stretching without tearing or gluing*), sometimes called *rubber-sheet geometry*. It is the *intrinsic character* of a space
- Topology is interesting for us, in robotics, because it affects our representation of space
- Topology allows us to use the same path in several spaces
- Two space have a different topology if cutting and pasting is required to make them the same – think of rubber figures – if we can stretch and reshape *continuously* without tearing, one shape/space into the other, then they have the same topology
- E.g., R^2 and T^2 are topologically different since no matter how T^2 is stretched without tearing there will always be a hole in it

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Topology

- A basic mathematical mechanism for talking about topology is the homeomorphism
- For a Mapping $\varphi: S \rightarrow T$
  - If each elements of φ goes to a unique T, φ is an *injection*
  - If each element of T has a corresponding *preimage* in S, then φ is a *surjection*
  - If φ is an *injection* and an *surjection*, then it is *bijection* (in which case an inverse, φ^{-1} exists)
  - φ is *smooth* if *derivatives of all orders exist* (we say φ *is* C^{∞})
- If $\varphi: S \rightarrow T$ is a *bijection*, and both φ and φ^{-1} are *continuous, φ is a homeomorphism;* if such a φ exists, S and T are *homeomorphic*
- If a *homeomorphism* where both φ and φ^{-1} are *smooth* is a *diffeomorphism*

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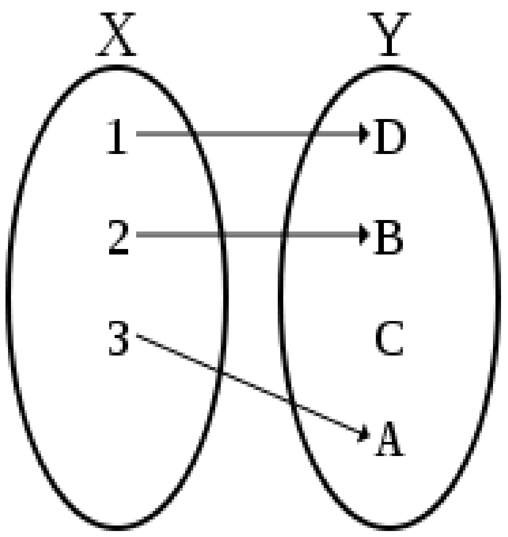
Topology : A lot of __jections

- *Injection*
  - In mathematics, an ***injective function*** is a function which associates distinct arguments with distinct values. An ***injective*** function is called an ***injection***, and is also said to be an ***information-preserving*** or ***one-to-one*** function



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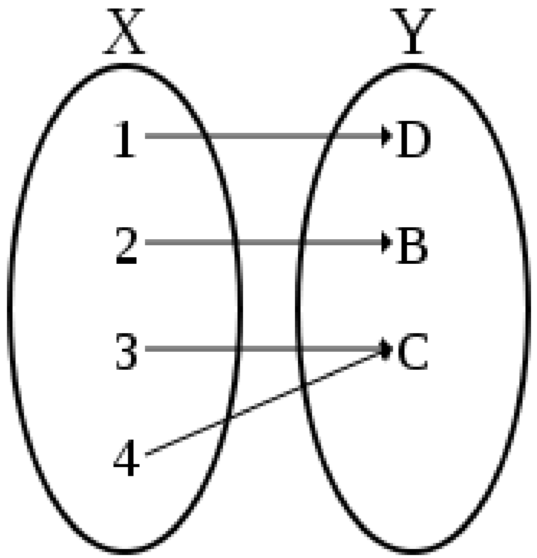
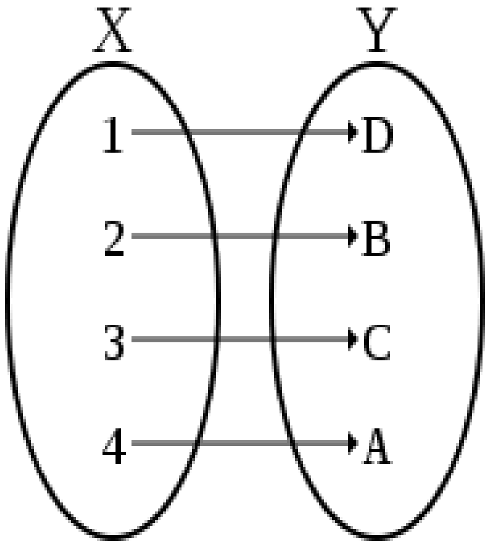
Topology : A lot of __jections

- ***Surjection***
  - In mathematics, a function *f* is said to be ***surjective*** or onto, if its values span its whole ***codomain***. A function $f: X \to Y$ is ***surjective*** if and only if its ***range*** $f(X)$ is equal to its ***codomain*** Y. A ***surjective*** function is called a ***surjection***



Topology : A lot of __jections

- ***Bijection***
  – ***f*** **is *bijective* if it is a *one-to-one correspondence* between those sets; i.e., both *one-to-one (injective)* and *onto (surjective)*.**



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Topology : A lot of __jections

- ***Continuous Functions***
  - A ***continuous*** process is one that proceeds without gaps of interruptions or sudden changes. Roughly speaking, a function $y = f(x)$ is continuous if it displays similar behaviour, that is, if a small change in x produces a small change in the corresponding value $f(x)$.

- ***Smooth Function***
  - The function f is said to be of class C^k if the derivatives f', f'', ..., $f^{(k)}$ exist and are ***continuous*** The function f is said to be of class C^∞, or smooth, if it has derivatives of all orders
    The function

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Configuration Space

Topology

- **If $\varphi: S \rightarrow T$ is a *bijection*, and both φ and φ^{-1} are *continuous*, φ *is a homeomorphism*; if such a φ exists, S and T are *homeomorphic***

- **If a *homeomorphism* where both φ and φ^{-1} are *smooth* is a *diffeomorphism***

A topologist cannot tell the difference between the cup in which he is drinking coffee and the doughnut that he is eating because